

ME597: Machine Learning (ML) Tutorial 1

Week 1

School of Mechanical Engineering
Purdue University

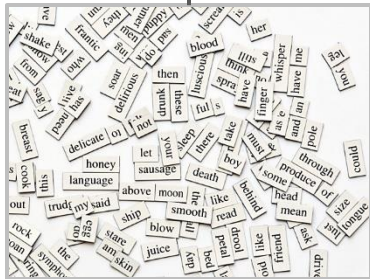
Dr. Martin Jun

- What is data?



- What is data?

*Everything
we can
Record*



Text Data



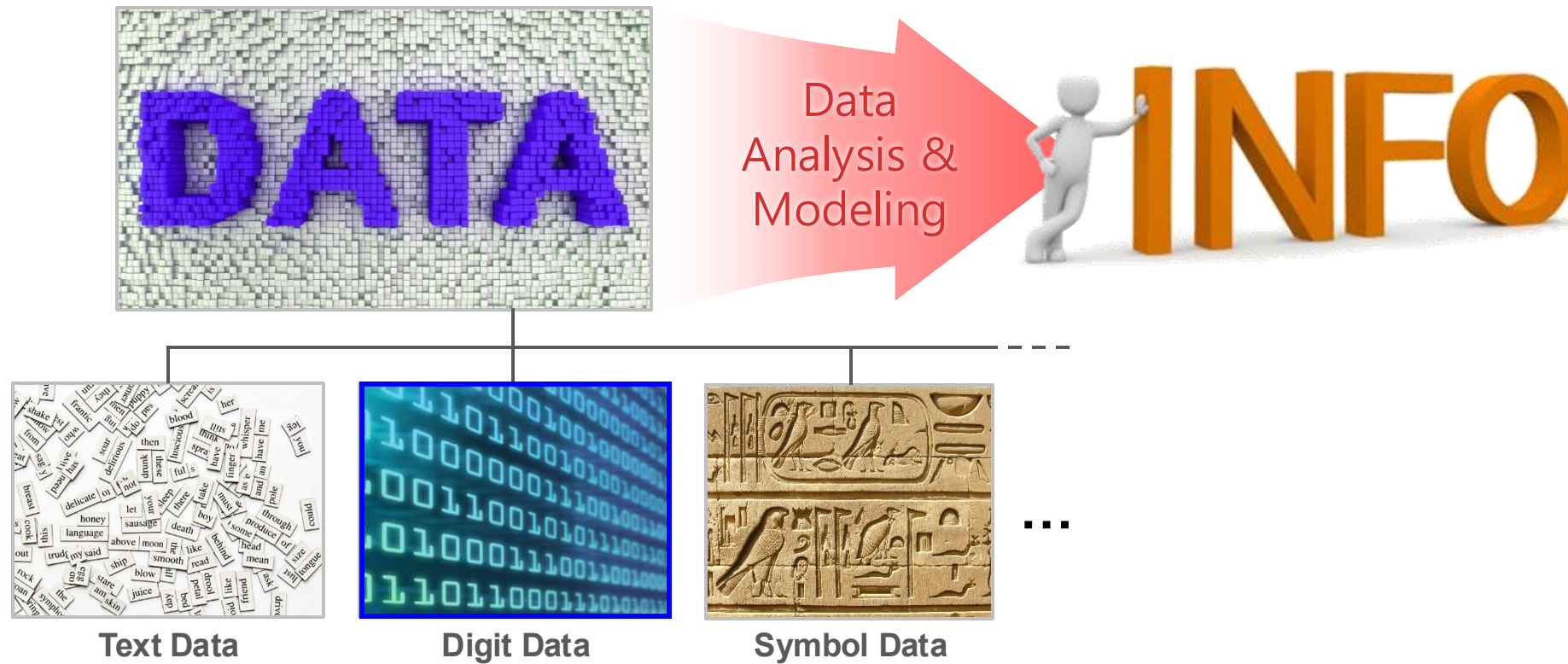
Digit Data



Symbol Data

...

- Data analysis & modeling (ML) for extracting meaning info. from data



- Type of digital data

* e.g., Temperature data

Scalar data

[e.g., Temperature measured
at a point in time]



36.6 °C

- Type of digital data

* e.g., Temperature data

Scalar data

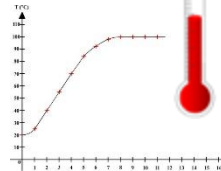
[e.g., Temperature measured at a point in time]



36.6 °C

Vector data

[e.g., Temperature measured continuously over a period of time]



9:30 am	36.5 °C
9:31 am	36.4 °C
9:32 am	36.7 °C
9:33 am	36.9 °C
9:34 am	37.1 °C
⋮	

- Type of digital data

* e.g., Temperature data

Scalar data

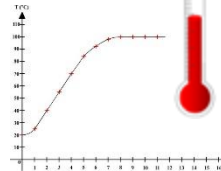
[e.g., Temperature measured at a point in time]



36.6 °C

Vector data

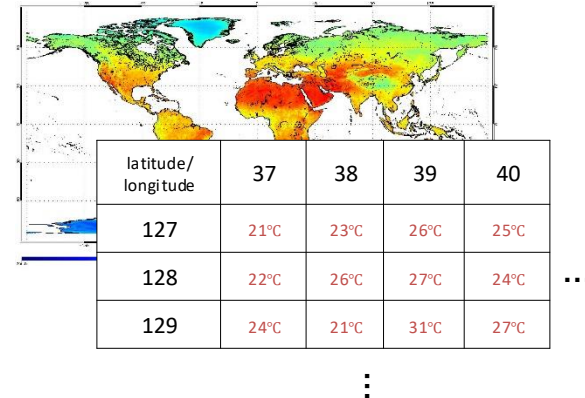
[e.g., Temperature measured continuously over a period of time]



9:30 am	36.5 °C
9:31 am	36.4 °C
9:32 am	36.7 °C
9:33 am	36.9 °C
9:34 am	37.1 °C
⋮	

Matrix data

[e.g., Temperature measured by location (2D)]

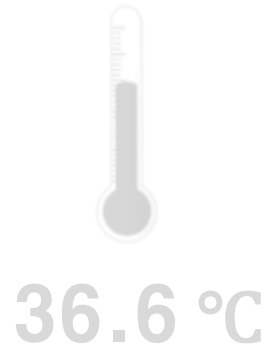


- Type of digital data

* e.g., Temperature data

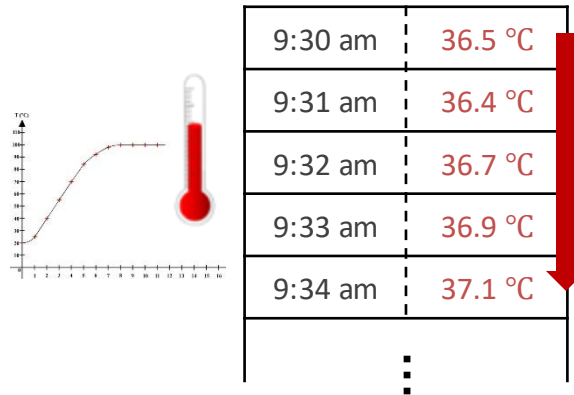
Scalar data

[e.g., Temperature measured at a point in time]



Vector data

[e.g., Temperature measured continuously over a period of time]



Sensors

Sensor type /
Bandwidth



Thermocouple
~ 100 Hz



Vision camera
~ 100 Hz(fps)



Force sensor
~ 10 kHz



Accelerometer
~ 20 kHz



Acoustic emission
~100 kHz



Ultrasonic sensor
~100 MHz

Collecting
small amount
& *long period*

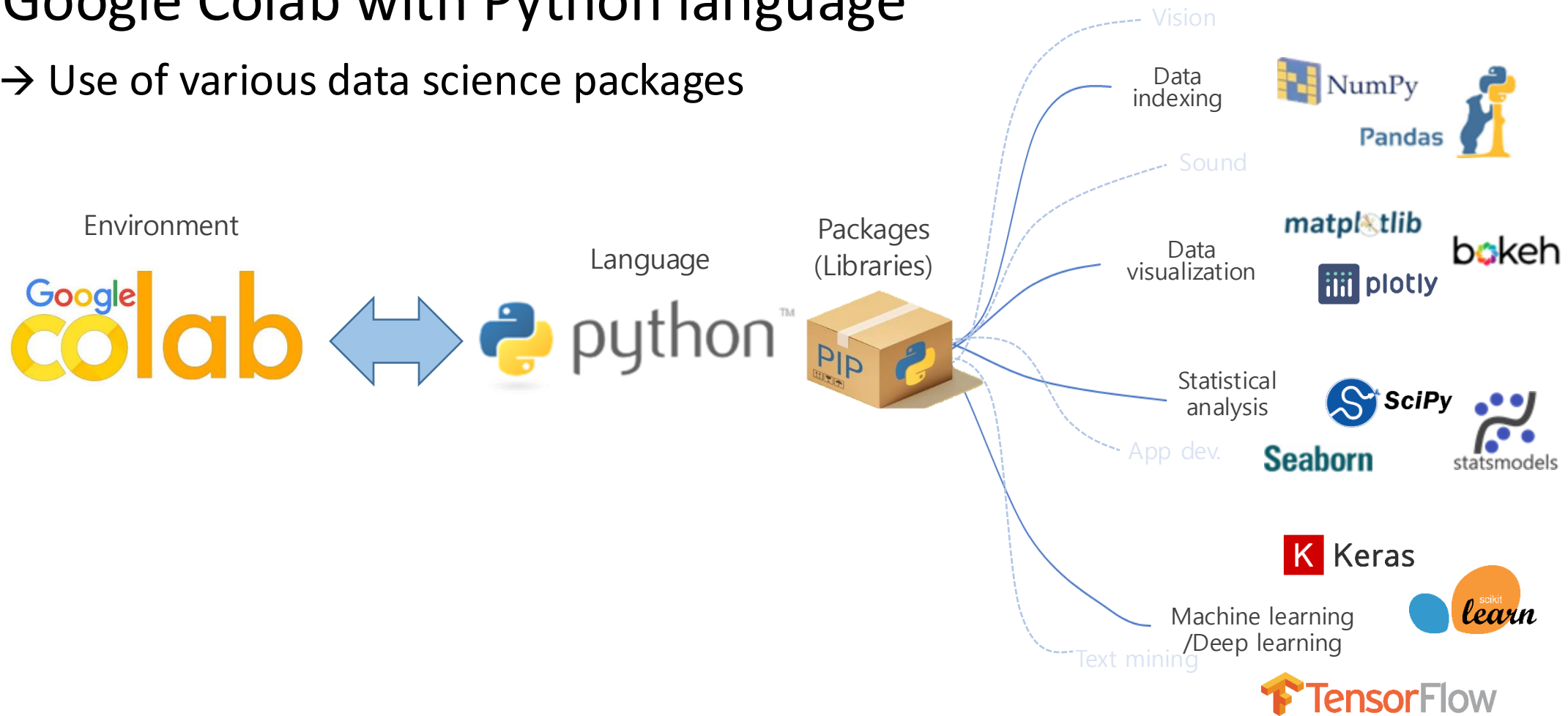


Collecting
large amount
& *short period*

Data Analysis & Modeling (ML) Environment

- Google Colab with Python language

→ Use of various data science packages



- Google Colab with Python language

- Use of various functions for data analysis and machine learning
- Basic structure of functions:

Package . Function (Parameter 1 , Parameter 2 , ...)

e.g., `sklearn.PCA( Data , dim. , param3)`: Principal Component Analysis (PCA)

- Output the result of function based on input values (parameters)
- What we need to do for using a function are:
 - (1) To know **what parameters are required** (understanding the concept of corresponding technique)
 - (2) To know **where each parameter should be input** (Google "package name + function name")

- Data format for data science in python

Data frame



- Data format of 'Pandas' package
 - Same as the data frame of 'R' (specialized in statistical analysis)
- Supports only vector (1D) and matrix (2D)

	time	Accelerometer	Voltage	Current	
	0	0.000000	0.016440	-0.014200	0.001433
	1	0.000078	0.013822	-0.015707	0.001504
	2	0.000156	0.019431	-0.016963	0.001433
	3	0.000234	0.003726	-0.017248	0.001549
	4	0.000313	0.022049	-0.009840	0.001864
...	...	...	...	...	...

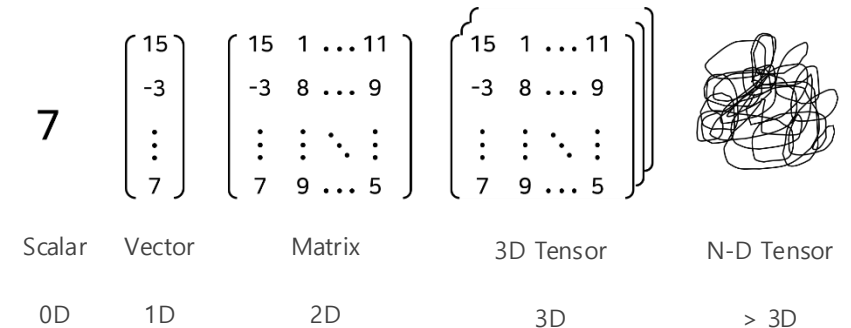
← index

→ To *load, save, and confirm data*

Array



- Data format of 'Numpy' package
 - An improved form of 'List' (basic array format) for easy operation
- Can be configured in various dimensions



→ To *process, analysis, and train (ML) data*